

UNDERGROUND SERVICES TOOLBOX TALK

The actual position of buried services may differ from drawings.

PURPOSE AND SCOPE

Prevent damage to buried electrical, gas, water, process and communication services during excavation, drilling, breaking and ground penetration.

KEY HAZARDS

1  Mechanical equipment striking live electrical	2  Damage to gas or hydrocarbon lines.	3  Uncontrolled release from pressurised water	4  Damage to communication or
5  Incorrect or outdated underground drawings.	6  Incorrect use of locating equipment.	7  Mechanical excavation too close to identified	8  Loss of surface markings during ongoing site

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Mechanical equipment striking live electrical cables.	02 Damage to gas or hydrocarbon lines.	03 Uncontrolled release from pressurised water or process
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REMEMBER — SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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FIELD REQUIREMENTS AND VERIFICATION

Prevent damage to buried electrical, gas, water, process and communication services during excavation, drilling, breaking and ground penetration.

SERNEM FIELD 01
FIELD VERIFICATION 1
Excavation permit available

SERNEM FIELD 02
FIELD VERIFICATION 2
Drawings reviewed

SERNEM FIELD 03
FIELD VERIFICATION 3
Area scanned

SERNEM FIELD 04
FIELD VERIFICATION 4
Services marked

VERIFICATION BEFORE START

- | | |
|--|--|
| ☐ Excavation permit available | ☐ Drawings reviewed |
| ☐ Area scanned | ☐ Services marked |
| ☐ Trial pit assessed | ☐ Safe distance defined |

SUPERVISOR DISCUSSION POINTS

- Do we have current utility drawings?
- Who scanned the area?
- Are detected services marked?

FIELD APPLICATION



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CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Review current utility drawings and excavation permits. • Maintain the site's required safe distance for mechanical excavation. • Stop work when drawings and field findings do not agree.

2. ENGINEERING & PHYSICAL CONTROLS

Have the area scanned by a competent person using suitable detection equipment. • Prepare emergency response for nearby gas or process services. • Record newly discovered services after completion.

3. WORK METHOD & FIELD CONTROLS

Mark identified service routes clearly on the ground. • Support and mechanically protect exposed services.

4. PPE, VERIFICATION & SUPERVISION

Use trial pits or controlled hand excavation in uncertain areas. • Repeat scanning as excavation depth progresses where necessary.

CRITICAL WARNING



Team, a clear drawing does not prove clear ground. No excavator will start until scanning, marking and controlled verification have been completed.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|--|--|
| ☐ Excavation permit available | ☐ Drawings reviewed |
| ☐ Area scanned | ☐ Services marked |
| ☐ Trial pit assessed | ☐ Safe distance defined |
| ☐ Operator briefed | ☐ Emergency plan ready |
| ☐ Field discrepancy checked | ☐ Crew informed |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			